

Derivation of 4D Fisher Information Matrix for Event-Based Sensors

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This document details the step-by-step derivation of the 4D Fisher information matrix from first principles.

1. INTRODUCTION

Information theory is the mathematical study of quantifying, storing, and transmitting information. First developed by Claude Shannon in 1948[1]. Since his initial publication, the field has been widely developed and has found use outside of signal processing, including any discipline where information can be thought of as needing to be quantified, stored, and/or transferred.

When information is quantized, we consider a single unit of information, either a bit, in the case of a binary outcome, or a nat, in the case of a continuous one. The nuances of which to choose is discussed in section B. For now, suffice it to know that information comes in discrete packets. One useful feature of this so-called Fisher information (named for the statistician and major contributor to the field of information theory, Sir Ronald Aylmer Fisher), is what it says about estimation. Specifically, the Cramer-Rao lower bound (CRLB) states that the variance of any unbiased estimator $\hat{\theta}$ of parameters θ cannot be less than the reciprocal of the information provided by the same. This is expressed mathematically in Eq. (1).

$$\text{Var}(\hat{\theta}[\theta]) \geq \frac{1}{\mathbb{I}(\theta)}$$

(1) Cramer-Rao Lower Bound

This imposes a lower bound on the variance, or in other words, an upper bound on precision, for any unbiased estimator. The lower bound gives us an absolute best case anyone can hope to achieve when estimating the values θ under the condition that the probability distribution used for the estimator is consistent with that used in the derivation of the lower bound.

This paper will derive a full 4 dimensional (4D) Fisher information matrix for an estimator of object localization in space, time, and spectra.

A. Notational Primer

Information theory is rooted in probability and statistics. As such it borrows much of its notation from established notations of the same. It will be beneficial, therefore, before proceeding, to provide a primer on the notation that will be used throughout this paper. This is provided by table S1. For questions regarding notation, refer back to this table.

Key Distinctions in Notation

- **Semicolon vs. Conditioning Bar:** $p(x \mid y)$ denotes a conditional probability distribution where y is an observed random variable. Conversely, $p(x; \theta)$ highlights that θ is a deterministic parameter structuring the distribution of x , not a random variable itself (unless adopting a Bayesian framework where it is written as $p(x \mid \theta)$).
- **Likelihood vs. Probability Density:** While $p(x; \theta)$ evaluates the probability/density of outcome x given a fixed parameter θ , the likelihood function $L(\theta; x)$ evaluates how plausible parameter values θ are given fixed observed data x .

Table S1. Foundational Notation and Terms in Information Theory and Statistics

Notation / Term	Name / Description	Definition, Usage, & Context
<i>Random Variables and Probability Distribution Functions</i>		
X, Y, Z	Random Variables	Capital letters denote random variables mapping outcomes to values.
x, y, z	Realizations / Samples	Lowercase letters denote specific realized values of random variables ($x \in \mathcal{X}$).
$\mathcal{X}, \mathcal{Y}$	Alphabet / State Space	Sets representing all possible values a random variable can take.
PMF/ $p_X(x)$	Probability Mass Function	$p_X(x) = P(X = x)$ for discrete random variables; satisfies $\sum_{x \in \mathcal{X}} p_X(x) = 1$.
PDF/ $f_X(x)$	Probability Density Function	Density function for continuous random variables; satisfies $\int_{-\infty}^{\infty} f_X(x) dx = 1$.
$P(X = x, Y = y)$	Joint Distribution	Probability that X and Y simultaneously take values x and y .
$P(X = x \mid Y = y)$ or $p(x \mid y)$	Conditional Distribution	Probability density/mass of X given $Y = y$; defined as $\frac{p(x, y)}{p(y)}$.
$P(x; \theta)$	Parametric Distribution	Probability density/mass of x parameterized by θ (fixed parameter, not a random variable).
<i>Parameters, Likelihood, and Estimation</i>		
θ	True Parameter (Vector)	Underlying parameter governing the distribution $p(x; \theta)$ or model.
$\hat{\theta}$	Estimator / Estimate	Function of sample data used to estimate θ (e.g., $\hat{\theta}_{\text{MLE}}$).
$L(\theta; x)$	Likelihood Function	$L(\theta; x) = p(x; \theta)$ viewed as a function of parameter θ given observed data x .
$\ell(\theta; x)$	Log-Likelihood Function	$\ell(\theta; x) = \ln L(\theta; x) = \ln p(x; \theta)$; turns products into sums for tractability.
<i>Expectations and Information Functions</i>		
$\mathbb{E}_X[g(X)]$	Expected Value	$\sum_x g(x)p(x)$ (discrete) or $\int g(x)f(x)dx$ (continuous).
$i(x) = -\log p(x)$	Self-Information (Surprisal)	Measure of information gained upon observing event x (measured in bits or nats).

2. INFORMATION AND ENTROPY

Shannon quantified the concept of surprise in terms of the information a given result may provide. He called this surprise "entropy," borrowing the term from the concept of the same name in thermodynamics. In information theory, entropy $H_P(f)$ for alphabet $\mathcal{X}$ is given by Eq. (2) where $0 \ln 0 = 0$ by definition.

$$H_P(f) = - \sum_{x \in \mathcal{X}} P(f = x) \ln P(f = x)$$

(a) Discrete Alphabet

$$h(f) = - \int_{\mathcal{X}} p(f) \ln p(f) df$$

(b) Continuous Alphabet

(2) Shannon entropy definitions.

This definition of entropy has many useful properties which are outside the scope of this paper. For a full treatise on the topic, consider "Entropy and Information Theory" [2]. To proceed towards the object of this paper, the relationship between entropy and information must be understood.

A. Joint and Conditional Shannon Information

To quantify uncertainty across multiple dependent variables, Shannon entropy extends naturally to joint distributions. For two random variables X and Y with joint probability distribution $p(x, y)$, the joint Shannon entropy $H(X, Y)$ measures the total average uncertainty contained in the pair:

$$H(X, Y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \ln p(x, y) = -\mathbb{E}_{X,Y}[\ln p(X, Y)]$$

(a) Discrete Joint Entropy

$$h(X, Y) = - \int_{\mathcal{X}} \int_{\mathcal{Y}} p(x, y) \ln p(x, y) dx dy = -\mathbb{E}_{X,Y}[\ln p(X, Y)]$$

(b) Continuous Differential Joint Entropy

(3) Shannon joint entropy definitions.

When observing variable Y , the remaining residual uncertainty in X is quantified by the conditional entropy $H(X | Y)$:

$$H(X | Y) = -\mathbb{E}_{X,Y}[\ln p(X | Y)] = H(X, Y) - H(Y) \quad (4)$$

The reduction in uncertainty about X achieved by observing Y defines the Mutual Information $I(X; Y)$:

$$I(X; Y) = H(X) - H(X | Y) = \mathbb{E}_{X,Y} \left[\ln \frac{p(X, Y)}{p(X)p(Y)} \right] \quad (5)$$

Mutual information measures the absolute shared information between random variables. However, when estimation of a non-random, deterministic parameter vector θ is desired, Shannon's probabilistic framework transitions into parametric statistical information.

B. Bridging Shannon Information to Fisher Information

While Shannon entropy measures the average state of randomness or descriptive uncertainty inherent to a distribution $p(x)$, Fisher information $\mathcal{I}(\theta)$ measures the sensitivity of a parameter-dependent distribution $p(x; \theta)$ with respect to infinitesimal perturbations in the underlying deterministic parameter vector θ .

Consider a data measurement $x \sim p(x; \theta)$. The localized information contributed by a single observation about θ is captured by how rapidly the log-likelihood function changes with respect to θ . We define the score function $V(\theta; x)$ as the gradient of the log-likelihood:

$$V(\theta; x) = \nabla_{\theta} \ell(\theta; x) = \nabla_{\theta} \ln p(x; \theta) = \frac{\nabla_{\theta} p(x; \theta)}{p(x; \theta)} \quad (6)$$

Under standard regularity conditions, specifically allowing the order of integration and differentiation to be interchanged, the expected value of the score function with respect to $p(x; \theta)$ vanishes identically:

$$\begin{aligned} \mathbb{E}_X[V(\theta; X)] &= \int_{\mathcal{X}} \left(\frac{\nabla_{\theta} p(x; \theta)}{p(x; \theta)} \right) p(x; \theta) dx \\ &= \nabla_{\theta} \int_{\mathcal{X}} p(x; \theta) dx = \nabla_{\theta}(1) = \mathbf{0} \end{aligned} \quad (7)$$

Fisher Information $\mathcal{I}(\theta)$ is formally defined as the covariance (or second moment) of the score function:

$$\mathcal{I}(\theta) \triangleq \mathbb{E}_X \left[(\nabla_{\theta} \ln p(X; \theta)) (\nabla_{\theta} \ln p(X; \theta))^T \right] \quad (8)$$

By differentiating Eq. (7) once more with respect to θ , an equivalent and computationally convenient representation emerges in terms of the expected negative Hessian of the log-likelihood:

$$\mathcal{I}(\theta) = -\mathbb{E}_X \left[\nabla_{\theta}^2 \ln p(X; \theta) \right] \quad (9)$$

B.1. The Connection: Local Shannon Entropy Curvature

The explicit bridge between Shannon differential entropy $h(X)$ and Fisher Information lies in a local Taylor expansion of entropy under parametric shifts. Let $p(x; \theta)$ be a parametric distribution and $p(x; \theta + \delta)$ be a slightly perturbed state. The Kullback-Leibler (KL) divergence expresses the relative Shannon entropy or excess surprisal between two distributions. The KL divergence admits a second-order expansion around $\delta = \mathbf{0}$:

$$D_{\text{KL}}(p(x; \theta) \parallel p(x; \theta + \delta)) = \frac{1}{2} \delta^T \mathcal{I}(\theta) \delta + \mathcal{O}(\|\delta\|^3) \quad (10)$$

Thus, Fisher Information dictates the local curvature of Shannon entropy space (specifically, the Riemannian metric tensor on the manifold of probability distributions). Higher Fisher information corresponds to high curvature in log-likelihood, meaning small shifts in parameters drastically alter the distribution shape—allowing precise estimation.

3. CRAMÉR-RAO LOWER BOUND

Understanding Fisher information and its relationship to Shannon entropy, we are prepared to consider the Cramér-Rao Lower Bound. We begin with an unbiased estimator $\hat{\theta}(X)$. It is expected that the average of all possible realized values X should equal the true value it is estimating: θ . This is expressed mathematically in Eq. (11).

$$\mathbb{E}_X [\hat{\theta}(X)] = \theta$$

(11) The mathematical definition of an unbiased estimator

differentiating both sides with respect to θ yields:

$$\nabla_{\theta} \mathbb{E}_X [\hat{\theta}(X) - \theta] = \mathbb{E}_X \left[(\hat{\theta}(X) - \theta) \nabla_{\theta} \ln p(X; \theta)^T \right] = \mathbf{I} \quad (12)$$

where $\mathbf{I}$ is the identity matrix. Applying the Cauchy-Schwarz inequality[3] for random vectors to Eq. (12) directly establishes the matrix Cramér-Rao inequality Eq. (13) where the inequality $\mathbf{A} \geq \mathbf{B}$ denotes that $\mathbf{A} - \mathbf{B}$ is a positive semi-definite matrix. The Fisher Information Matrix $\mathcal{I}(\theta)$ therefore sets the fundamental physical limit on object parameter estimation precision.

$$\text{Var}(\hat{\theta}) \geq \mathcal{I}(\theta)^{-1}$$

(13) The Cramer-Rao Lower Bound

4. EVENT-BASED POINT SPREAD FUNCTIONS

Event-Based sensors (EBS) operate on fundamentally different physics principles than traditional digital image sensors. Instead of collecting photons for a fixed period across a uniform pixel grid, EBS pixels operate as independent, asynchronous detectors. Each pixel contains logarithmic front-end circuitry coupled to a differential comparator that continuously monitors temporal light variations. An event $e_i = (x_i, y_i, t_i, p_i)$ is triggered at microsecond resolution whenever the localized log-intensity change crosses a preset contrast threshold C :

$$\Delta \ln I(x_i, y_i, t_i) = \ln I(x_i, y_i, t_i) - \ln I(x_i, y_i, t_i - \Delta t) \geq \pm C \quad (14)$$

where $p_i \in \{-1, +1\}$ denotes the event polarity (brightness decrease or increase).

This architecture provides several key operational advantages:

- **Low Power Consumption:** Pixels remain quiescent until local scene radiance changes, eliminating redundant frame readouts.
- **High Dynamic Range (HDR):** Logarithmic compression allows operation across dynamic ranges exceeding 120 dB without over-exposure.
- **Sub-Millisecond Temporal Resolution:** Asynchronous detection yields microsecond-level timing precision.

To leverage these advantages for spatial, temporal, and spectral estimation, optical modeling must bridge continuous physical radiance fields with sparse, temporal point-process event streams.

A. Point Spread Functions in Computational Imaging

In classical imaging optics, the Point Spread Function (PSF) $h(x, y; \theta)$ defines the optical transfer function mapping a point source parameterized by θ (e.g., spatial location (u_0, v_0) , depth z , or wavelength λ) to an intensity distribution on the focal plane:

$$I(x, y; \theta) = |\mathcal{F}\{P(\rho, \phi) e^{i\psi(\rho, \phi; \theta)}\}|^2 \quad (15)$$

where $P(\rho, \phi)$ represents the pupil amplitude aperture and $\psi(\rho, \phi; \theta)$ denotes a phase profile introduced by optical elements or custom diffractive phase masks.

In computational imaging, the PSF is deliberately engineered rather than minimized for diffraction-limited focus. By engineering the phase profile $\psi(\rho, \phi)$, the optical system acts as a physical encoder, multiplexing higher-dimensional scene properties into spatial intensity distributions on a 2D sensor array:

- **Depth Encoding:** Rotating or astigmatic PSFs transform axial defocus into spatial orientation or feature scale, enabling single-shot 3D localization.
- **Spectral Discrimination:** Chromatic dispersion engineered through diffractive optical elements (DOEs) encodes spectral emission lines into spatial intensity patterns.
- **Velocity Sensing:** Directional optical blurs map motion vectors directly to spatial energy distributions.

B. PSF Engineering for Asynchronous Event Sensing

While phase mask design and PSF optimization are well-established for frame-based intensity integration, extending PSF engineering to event-based vision sensors remains an active area of investigation.

Conventional optical design metrics, such as minimizing the root-mean-square (RMS) spot size, maximizing the Strehl ratio, or optimizing spatial modulation transfer functions (MTF), are tailored for static energy accumulation on time-integrating CMOS sensors. However, as

established in Eq. (14), event cameras are entirely blind to static intensity distributions; events are generated exclusively by spatio-temporal log-intensity derivatives Eq. (16).

$$\frac{\partial}{\partial t} \ln I(x, y, t; \theta) = \frac{1}{I(x, y, t; \theta)} \left(\nabla_{(x, y)} I \cdot \mathbf{v} + \frac{\partial I}{\partial t} \right)$$

(16) EBS contrast response forward model

where $\mathbf{v} = (v_x, v_y)^T$ is the transverse velocity vector of the source.

Consequently, an optimal event-based PSF must not maximize spatial brightness concentration, but rather maximize the high-frequency spatial-temporal gradients produced under object motion. Designing diffractive phase profiles $\psi(\rho, \phi)$ specifically tailored to maximize the score function $\nabla_{\theta} \ln p(\text{events}; \theta)$ and the resulting Fisher Information Matrix $\mathcal{I}(\theta)$ under sparse Poisson event dynamics represents a critical open challenge in modern computational optics.

One path toward an optimal event-based PSF is machine learning. As demonstrated by Shah et al. [4], the CRLB offers a well-tuned objective for a machine learning algorithm to optimize for. In their work, the intensity response was approximated as a Gaussian of the intensity differences of pseudo-frames (i.e., accumulated and binned events over the sensor area). This works well, provided that the integration period is sufficiently long for the approximation to apply. However, it will be shown that this requirement creates an adverse incentive, as information degrades with accumulation time in EBS. Furthermore, in their model, time is an implicit parameter, and velocity is not considered. Here we expand on the former work by deriving a CRLB in time, space, and spectra for an EBS.

5. DERIVATION

We begin first by considering CRLB of a Poisson distribution parameterized over a single variable. The choice of a Poisson distribution is to accurately model the asynchronous and independent nature of the events.

A. Poisson CRLB for Parameterized Rates

To model the asynchronous events generated by the sensor, we observe the number of events K occurring within a given time interval Δt . Because event generation is driven by photon arrival times at the photodiode, we model this as a Poisson process [5]. The probability mass function for observing k events given a rate parameter Λ is:

$$f(k; \Lambda) = \frac{(\Lambda \Delta t)^k e^{-\Lambda \Delta t}}{k!} \quad (17)$$

To find the Fisher information of this process, we first calculate the log-likelihood $L = \ln f(k; \Lambda)$:

$$\begin{aligned} L &= \ln \left((\Lambda \Delta t)^k \right) + \ln \left(e^{-\Lambda \Delta t} \right) - \ln(k!) \\ &= k \ln(\Lambda) + k \ln(\Delta t) - \Lambda \Delta t - \ln(k!) \end{aligned} \quad (18)$$

The score function S_{Λ} is the first partial derivative of the log-likelihood with respect to the rate parameter Λ . Since the terms $k \ln(\Delta t)$ and $\ln(k!)$ do not depend on Λ , their derivatives vanish:

$$S_{\Lambda} = \frac{\partial L}{\partial \Lambda} = \frac{k}{\Lambda} - \Delta t$$

(19) Score function of the Poisson event count

The Fisher information $\mathcal{I}(\Lambda)$ is defined as the expected value of the squared score function, or equivalently, the negative expected value of the second derivative of the log-likelihood. Taking the second derivative yields:

$$\frac{\partial^2 L}{\partial \Lambda^2} = -\frac{k}{\Lambda^2} \quad (20)$$

Taking the negative expectation and recognizing that the expected value of a Poisson random variable K over interval Δt is $\mathbb{E}[K] = \Lambda \Delta t$:

$$\mathcal{I}(\Lambda) = -\mathbb{E} \left[-\frac{K}{\Lambda^2} \right] = \frac{\Lambda \Delta t}{\Lambda^2} = \frac{\Delta t}{\Lambda}$$

(21) Fisher information of the Poisson rate parameter

B. Physical Forward Model and the Chain Rule

In an event camera, the local event rate $\Lambda(u, v, t)$ is directly proportional to the magnitude of the continuous log-intensity time derivative, plus a baseline background noise rate Λ_0 :

$$\Lambda(u, v, t; \theta) = C \left| \frac{\partial \ln I(u, v, t; \theta)}{\partial t} \right| + \Lambda_0 = \frac{C}{I} \left| \frac{\partial I}{\partial t} \right| + \Lambda_0 \quad (22)$$

where C is a scaling sensitivity parameter linked to the sensor's contrast threshold.

To determine the Fisher information with respect to an arbitrary physical parameter θ_i , we apply the chain rule. The score function for θ_i is $S_{\theta_i} = S_{\Lambda} \frac{\partial \Lambda}{\partial \theta_i}$. Because I and its temporal derivative both depend on θ_i , we must use the product rule when differentiating the rate:

$$\frac{\partial \Lambda}{\partial \theta_i} = C \left[\frac{1}{I} \frac{\partial^2 I}{\partial \theta_i \partial t} - \frac{1}{I^2} \frac{\partial I}{\partial \theta_i} \frac{\partial I}{\partial t} \right]$$

(23) Partial derivative of the event rate with respect to an arbitrary optical parameter applying the product rule

assuming the intensity is increasing to drop the absolute value without loss of generality.

The Fisher information for θ_i at a single pixel is therefore:

$$\mathcal{I}(\theta_i) = \mathbb{E}[S_{\theta_i}^2] = \left(\frac{\partial \Lambda}{\partial \theta_i} \right)^2 \mathbb{E}[S_{\Lambda}^2] \quad (24)$$

Substituting Eq. (21) and Eq. (23) yields the Fisher information for a single independent pixel:

$$\mathcal{I}(\theta_i) = \frac{\Delta t}{\Lambda} \left(\frac{C}{I} \left[\frac{\partial^2 I}{\partial \theta_i \partial t} - \frac{1}{I} \frac{\partial I}{\partial \theta_i} \frac{\partial I}{\partial t} \right] \right)^2 \quad (25)$$

Because event generation at each pixel (u, v) is statistically independent, the total Fisher information for the entire sensor array is additive:

$$\mathcal{I}_{total}(\theta_i) = \sum_{u,v} \frac{\Delta t}{\Lambda_{u,v}} \left(\frac{C}{I_{u,v}} \left[\frac{\partial^2 I_{u,v}}{\partial \theta_i \partial t} - \frac{1}{I_{u,v}} \frac{\partial I_{u,v}}{\partial \theta_i} \frac{\partial I_{u,v}}{\partial t} \right] \right)^2$$

(26) Total additive Fisher information for the entire event sensor array

C. Joint Parameter Estimation and System Matrix

To fully capture the behavior of the optical system, the parameters must be grouped into dynamic source parameters θ_s and static optical parameters θ_h . The full joint parameter vector is defined as $\theta = [\theta_s, \theta_h]^T$, where:

- $\theta_s = \{v_u, v_v, \nabla I, \Delta \lambda, \Delta I\}$ dictates the source motion and spectral shifts driving the temporal event triggers.
- $\theta_h = \{\phi, u_0, v_0, I_0\}$ defines the static optical layout and phase mask parameters generating the PSF.

Estimating these jointly in a Fisher Information Matrix $\mathcal{I}(\theta)$ evaluates the cross-correlations (off-diagonal elements), explicitly modeling how spectral shifts ($\Delta \lambda$) deform the PSF through chromatic dispersion in the phase mask (ϕ).

C.1. Expanded Matrix Representation

The fully expanded 9×9 Fisher Information Matrix explicitly details the coupling across all dynamic parameters $\theta_s = [v_u, v_v, \nabla I, \Delta\lambda, \Delta I]^T$ and static optical parameters $\theta_h = [\phi, u_0, v_0, I_0]^T$:

$$\mathcal{I}(\theta) = \begin{bmatrix} \mathcal{I}_{v_u, v_u} & \mathcal{I}_{v_u, v_v} & \cdots & \mathcal{I}_{v_u, \Delta\lambda} & \cdots & \mathcal{I}_{v_u, \phi} & \cdots & \mathcal{I}_{v_u, I_0} \\ \mathcal{I}_{v_v, v_u} & \mathcal{I}_{v_v, v_v} & \cdots & \mathcal{I}_{v_v, \Delta\lambda} & \cdots & \mathcal{I}_{v_v, \phi} & \cdots & \mathcal{I}_{v_v, I_0} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \mathcal{I}_{\Delta\lambda, v_u} & \mathcal{I}_{\Delta\lambda, v_v} & \cdots & \mathcal{I}_{\Delta\lambda, \Delta\lambda} & \cdots & \mathcal{I}_{\Delta\lambda, \phi} & \cdots & \mathcal{I}_{\Delta\lambda, I_0} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \mathcal{I}_{\phi, v_u} & \mathcal{I}_{\phi, v_v} & \cdots & \mathcal{I}_{\phi, \Delta\lambda} & \cdots & \mathcal{I}_{\phi, \phi} & \cdots & \mathcal{I}_{\phi, I_0} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \mathcal{I}_{I_0, v_u} & \mathcal{I}_{I_0, v_v} & \cdots & \mathcal{I}_{I_0, \Delta\lambda} & \cdots & \mathcal{I}_{I_0, \phi} & \cdots & \mathcal{I}_{I_0, I_0} \end{bmatrix}$$

(27) Expanded 4D Fisher Information Matrix highlighting the cross-talk term $\mathcal{I}_{\Delta\lambda, \phi}$ between spectral shift and phase mask geometry

The off-diagonal terms $\mathcal{I}_{\Delta\lambda, \phi}$ and $\mathcal{I}_{\phi, \Delta\lambda}$ quantify the cross-correlation introduced as spectral variations alter the phase shift profile through optical dispersion.

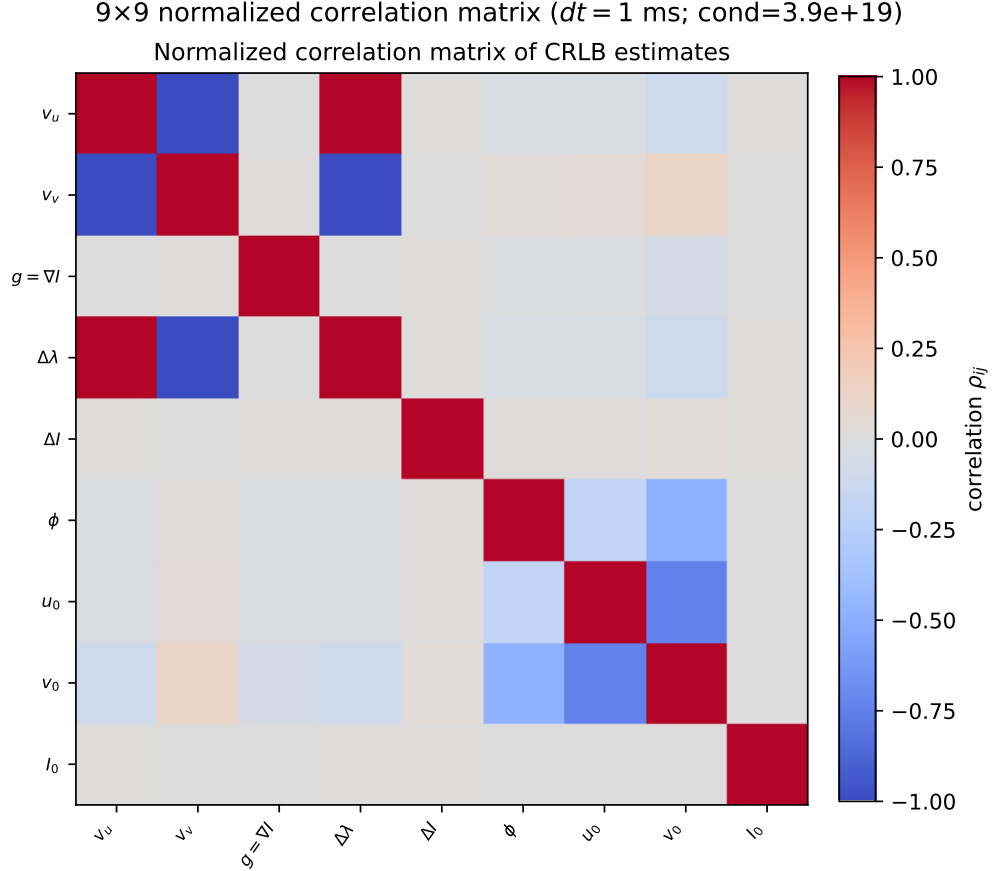


Fig. S1. Normalized covariance of the 9x9 FIM under a simplified PSF model.

C.2. Condensed Block Matrix Representation

Partitioning the parameter vector into dynamic (θ_s) and static (θ_h) sub-vectors allows the system matrix to be expressed in compact block form:

$$\mathcal{I}(\theta) = \begin{bmatrix} \mathcal{I}_{\theta_s, \theta_s} & \mathcal{I}_{\theta_s, \theta_h} \\ \mathcal{I}_{\theta_h, \theta_s} & \mathcal{I}_{\theta_h, \theta_h} \end{bmatrix}$$

(28) Condensed block structure of the joint Fisher Information Matrix

where $\mathcal{I}_{\theta_s, \theta_s} \in \mathbb{R}^{5 \times 5}$ represents internal dynamic correlations, $\mathcal{I}_{\theta_h, \theta_h} \in \mathbb{R}^{4 \times 4}$ represents static layout correlations, and $\mathcal{I}_{\theta_s, \theta_h} = \mathcal{I}_{\theta_h, \theta_s}^T \in \mathbb{R}^{5 \times 4}$ encapsulates all cross-coupling mechanisms between dynamic source variations and optical mask design.

D. Performance Scaling with Illumination

A fundamental physical limitation is revealed when analyzing this bound as a function of total photon count N . If we decompose total intensity as $I_{u,v} = N \cdot \hat{I}_{u,v}$, where $\hat{I}_{u,v}$ represents the normalized spatial shape, the constant N cancels entirely out of the log-intensity time derivative:

$$\frac{1}{N \hat{I}_{u,v}} \frac{\partial(N \hat{I}_{u,v})}{\partial t} = \frac{1}{\hat{I}_{u,v}} \frac{\partial \hat{I}_{u,v}}{\partial t} \quad (29)$$

The signal generation is purely dependent on relative contrast and invariant to absolute brightness. However, the background rate $\Lambda_0(N)$ grows with the total photon count. Applying this to Eq. (25), the Fisher information is proportional to the inverse of the total rate:

$$\mathcal{I}(\theta_i) \propto \frac{1}{\Lambda_{\text{signal}} + \Lambda_0(N)} \quad (30)$$

Because the CRLB is the inverse of the Fisher information, the theoretical error bound for an event sensor grows linearly with N . In contrast to standard frame-based cameras where information scales favorably with absolute light, an event camera's optimal performance regime exists at lower relative illumination before ambient background shot noise dominates the contrast signal.

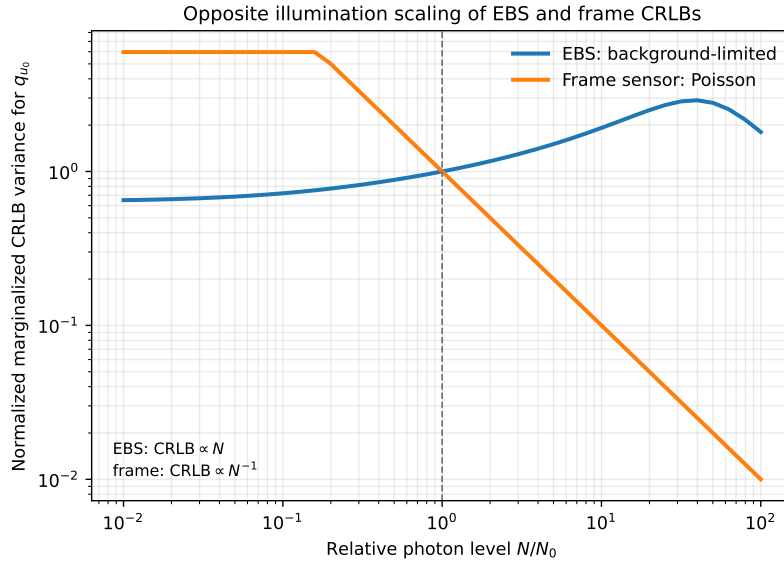


Fig. S2. The normalized CRLB of the event sensor compared with the frame sensor.

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